

Aegis: A Latent World Model as a Margin-of-Stability Safety Critic for Fall-Risk-Aware Lower-Limb Exoskeleton Assistance

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Abstract

Falls during locomotion are a leading cause of injury among older and mobility-impaired adults, and assistive lower-limb exoskeletons that inject joint torque can themselves perturb balance when assistance is mistimed. We present *Aegis*, an action-conditioned latent world model that acts as a *margin-of-stability safety critic* for a bilateral lower-limb exoskeleton with six actuated joints (hip, knee, ankle) and a pelvis-mounted reaction-wheel momentum module. Aegis encodes proprioceptive, exteroceptive (egocentric terrain heightmap), and centroidal state into a latent space, predicts future centroidal trajectories conditioned on candidate assistance torques, and decodes a learned margin of stability defined through the extrapolated center of mass. At deployment the critic admits only assistance actions whose predicted margin remains above a buffer over a short horizon, projects any unsafe proposal from a higher-level policy onto this safe set, and keeps a certified low-level impedance controller in the loop at all times. We prove that, under a bounded multi-step prediction error, projection onto the predicted-safe set keeps the *true* extrapolated center of mass within the support polygon over the horizon, yielding a falsifiable margin guarantee (Prop. 1); and we show through a reduced-order frontal-plane analysis that the reaction wheel strictly enlarges the set of states from which a safe assistance action exists—non-empty precisely in the single-support regime where a foot-only device loses frontal-plane authority (Prop. 2). We detail the architecture, a boundary-weighted training objective, a conformal calibration of the safety buffer, and a complete evaluation protocol spanning simulated perturbation studies and a hardware validation plan. We position Aegis as the world-model substrate for a subsequent language-conditioned vision-language-action (VLA) policy that reuses the same critic as a safety filter.

1 Introduction

Falls are among the most consequential health risks for older adults and people with neuromuscular impairment, with the majority occurring during walking and balance transitions [1]. Powered lower-limb exoskeletons can offset metabolic cost and augment joint power [9, 10], and a growing body of work optimizes *when* and *how much* torque to deliver [10, 11]. Yet most assistive controllers are either *reactive*—they respond to a perturbation only after it registers in proprioception—or *blind*—they ignore the terrain and the body’s centroidal state. This is precisely the regime in which a device can do harm: a well-intended flexion assist applied at the wrong instant can push the body’s center of mass beyond its base of support and *cause* the fall it was meant to prevent.

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We argue that fall-risk-aware assistance should be posed as *constrained action selection against a predictive model of balance*. Concretely: before any assistance torque is applied, a model should predict the body’s future centroidal trajectory *under that candidate torque* and verify that a principled balance criterion is not violated over a short horizon. This is the role we give to a *world model* [12, 14, 15]: not to generate motion, but to serve as a fast, action-conditioned *safety critic* that vetoes or repairs unsafe assistance.

Two design choices make this tractable and rigorous. First, we ground the critic in the *margin of stability* (MoS) defined through the extrapolated center of mass (XCoM) [2, 3], a biomechanically standard, scalar balance criterion with a clear geometric meaning—the signed distance from the velocity-extrapolated CoM to the edge of the support polygon. Predicting this margin, rather than reconstructing full state, concentrates learning on the one quantity that determines whether the body falls. Second, the device carries a *pelvis-mounted reaction-wheel momentum module* in addition to its six actuated leg joints. This is not incidental: in single support a foot-only device has bounded frontal-plane authority (the ankle/CoP “strategy” is limited by foot width), and there exist states from which *no* admissible leg torque can arrest a lateral loss of balance. A reaction wheel injects a frontal-plane moment without a ground-contact constraint [8, 5], and we show it strictly enlarges the recoverable set in exactly that regime.

Figure 1 shows the device. Our contributions are:

1. **Formulation.** We cast exoskeleton balance assistance as constrained action selection against a learned, predictive margin of stability, and instantiate it with an action-conditioned latent world model (Aegis) that fuses proprioception, egocentric terrain, and centroidal state (Sec. 4).
2. **Learned margin head.** We train a stability-margin head grounded in the XCoM/MoS criterion with a *boundary-weighted* loss so accuracy concentrates near $b=0$, the only region that matters for safety (Sec. 4.5).
3. **Safety-critic deployment with a guarantee.** The critic projects higher-level proposals onto the predicted-safe set while a certified low-level controller stays in the loop. We prove a margin guarantee under bounded prediction error (Prop. 1) and make it operational via split-conformal calibration of the buffer (Cor. 1).
4. **Hardware-grounded controllability result.** A reduced-order frontal-plane analysis proves the reaction wheel strictly enlarges the feasible safe set in single support (Prop. 2), tying a hardware design choice to the controllability of balance (Sec. 5).
5. **Evaluation protocol.** We give a complete simulation-plus-hardware evaluation plan with baselines and ablations (Sec. 6), and position Aegis as the substrate for a downstream language-conditioned VLA policy (Sec. 7).

2 Related Work

Exoskeleton assistance. Unpowered and powered exoskeletons reduce the metabolic cost of walking and running [9], and human-in-the-loop optimization tunes assistance to the wearer [10]. Real-time gait-phase estimation enables phase-based torque profiles on hip devices [11]. These methods optimize efficiency or tracking; they do not maintain an explicit, predictive balance constraint, which is our focus.

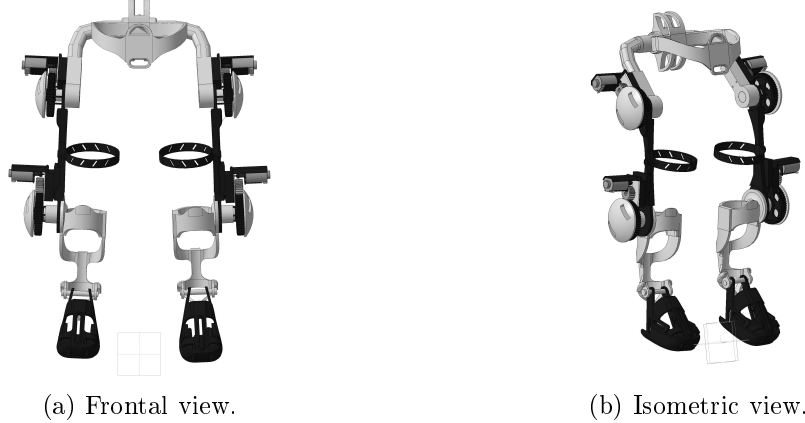


Figure 1: The bilateral lower-limb exoskeleton analyzed in this work. Six actuated joints—hip, knee, and ankle on each side—drive the leg linkages, while a pelvis-mounted reaction-wheel module (top bridge) provides frontal-plane moment authority that does not depend on ground contact. The actuation channels define the seven-dimensional assistance action of Sec. 3.2.

Balance criteria and push recovery. The linear inverted pendulum model (LIPM) [4], the capture point [5], and capturability analysis [6] formalize when a legged system can come to a stop, and trajectory-free MPC stabilizes walking under perturbations [7]. The margin of stability via the extrapolated CoM [2, 3] is the biomechanics-standard scalar we adopt. The reaction-mass pendulum [8] models centroidal angular momentum and motivates flywheel/CMG-style recovery; we use it to analyze the pelvis wheel.

World models and model-based control. World models learn action-conditioned latent dynamics for prediction and control [12, 13, 14, 15], descending from Dyna-style model-based RL [16]. We do not use the model to imagine policies; we use it as a one-step-ahead *safety critic*.

Safe learning and control. Safe RL constrains policies during learning and deployment [21, 20]; model-based safe RL provides stability guarantees under Lipschitz/Gaussian-process assumptions [17]; and control barrier functions render a safe set forward-invariant for known dynamics [18, 19]. Our margin head plays a role analogous to a barrier function, but it is *learned* and *predictive*, and our guarantee is conditioned on a calibrated bound on model error obtained by conformal prediction [22].

Vision-language-action policies. VLA models map instructions and observations to robot actions [23, 24, 25]. Aegis is the world-model layer beneath such a policy: the VLA proposes *what* to do; the critic guarantees the proposal will not violate balance (Sec. 7).

3 Problem Formulation

3.1 Device and actuation

The exoskeleton (Fig. 1) is a bilateral lower-limb device with hip, knee, and ankle actuation per side (six actuated joint degrees of freedom) and a pelvis-mounted reaction-wheel module that produces a frontal-plane moment. The wheel is a *transient* actuator: it can inject angular momentum

quickly but saturates in wheel speed, making it suited to brief stumble-recovery windows rather than sustained support—exactly the horizon over which a fall is decided.

3.2 State and action spaces

Let the observed state at time t be

$$s_t = \left[\underbrace{q, \dot{q}}_{\text{joints}}, \underbrace{p_c, \dot{p}_c}_{\text{CoM}}, \underbrace{\theta_{\text{pel}}, \omega_{\text{pel}}}_{\text{pelvis}}, \underbrace{c_L, c_R}_{\text{contacts}}, \underbrace{H_t}_{\text{terrain}}, \underbrace{\phi_{\text{rw}}, \dot{\phi}_{\text{rw}}}_{\text{wheel}} \right], \quad (1)$$

where $q, \dot{q} \in \mathbb{R}^6$ are joint positions and velocities (encoder-derived); $p_c \in \mathbb{R}^3$ is the whole-body CoM position estimated from kinematics and IMU fusion, with velocity $\dot{p}_c$; $\theta_{\text{pel}}, \omega_{\text{pel}}$ are pelvis orientation and angular rate; $c_L, c_R \in [0, 1]$ are (soft) foot-contact indicators that, together with foot geometry, define the support polygon $\mathcal{S}_t \subset \mathbb{R}^2$ in the ground plane; H_t is a local terrain heightmap (a 32×32 patch ahead of the swing foot) from an egocentric depth camera; and $\phi_{\text{rw}}, \dot{\phi}_{\text{rw}}$ are the reaction-wheel angle and speed.

The assistance action is the vector of joint and wheel torques (applied as deltas over a nominal phase-based controller; see Sec. 4.6):

$$a_t = [\tau_{\text{hip}}^L, \tau_{\text{hip}}^R, \tau_{\text{knee}}^L, \tau_{\text{knee}}^R, \tau_{\text{ank}}^L, \tau_{\text{ank}}^R, \tau_{\text{rw}}] \in \mathbb{R}^7. \quad (2)$$

The reaction-wheel torque τ_{rw} is the channel that produces a frontal-plane moment on the pelvis *without* a ground-contact constraint and is therefore available throughout single support.

3.3 Balance: the margin of stability

We measure balance with the extrapolated center of mass (XCoM). Let p_c^{xy} and $\dot{p}_c^{xy}$ be the ground-plane CoM position and velocity, ℓ the CoM height, g gravity, and $\omega_0 = \sqrt{g/\ell}$. The XCoM is

$$\xi_t = p_c^{xy} + \frac{\dot{p}_c^{xy}}{\omega_0}, \quad (3)$$

and the (signed) margin of stability is its distance to the nearest edge of the support polygon,

$$b_t = \min_{e \in \partial \mathcal{S}_t} \text{dist}(\xi_t, e), \quad (4)$$

with the convention $b_t > 0$ when ξ_t lies inside $\mathcal{S}_t$ (dynamically balanced) and $b_t \leq 0$ when it has left the support polygon, at which point a step or an external moment is *required* to avoid falling [2, 3, 5]. The quantity the safety critic predicts and protects is the *future* margin b_{t+k} under a candidate assistance action.

4 Method: Aegis

Aegis has three learned components—an encoder, an action-conditioned latent dynamics model, and a decoder with a dedicated stability-margin head—and one non-learned deployment rule, the safety critic. Figure 2 shows the architecture; Figure 3 places it in the control hierarchy.

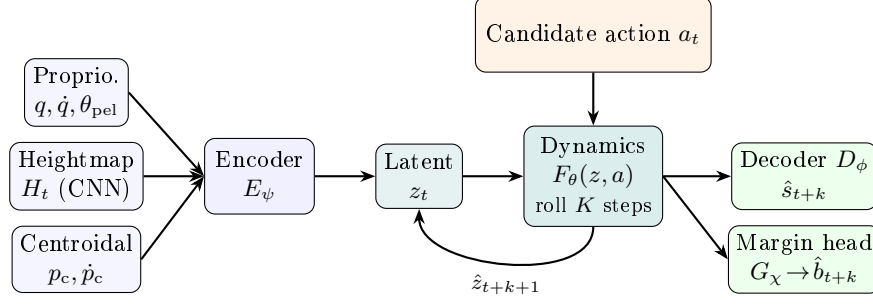


Figure 2: Aegis architecture. Proprioceptive, exteroceptive (terrain heightmap, via a small CNN), and centroidal inputs are encoded to a latent z_t . The action-conditioned latent dynamics F_θ roll forward K steps given a candidate assistance action; the decoder reconstructs predicted state and a dedicated margin head G_χ outputs the predicted margin of stability $\hat{b}_{t+k}$ directly.

4.1 Action-conditioned latent world model

The encoder maps an observation to a latent; the dynamics predict the next latent conditioned on the action; the decoder reconstructs state:

$$z_t = E_\psi(s_t), \quad \hat{z}_{t+1} = F_\theta(z_t, a_t), \quad \hat{s}_{t+k} = D_\phi(\hat{z}_{t+k}). \quad (5)$$

Rolling (5) forward autoregressively for $k = 1, \dots, K$ yields a predicted trajectory $\{\hat{s}_{t+k}\}_{k=1}^K$ over horizon $T_H = K \Delta t$ (target $T_H \approx 0.5\text{--}1.0\text{s}$, covering a stumble-recovery window). We use a deterministic residual latent transition for the main result—determinism makes the critic’s verdict reproducible—and discuss a stochastic state-space variant in Sec. 7.

4.2 Stability-margin head

Rather than computing the margin only by decoding state and applying (3)–(4) post hoc, we attach a head G_χ that predicts the margin directly from the latent,

$$\hat{b}_{t+k} = G_\chi(\hat{z}_{t+k}), \quad (6)$$

supervised by the geometric margin b_{t+k} computed from ground-truth state. Making the margin a first-class learned output, and weighting its loss toward the boundary $b \approx 0$ (Sec. 4.5), concentrates capacity where safety is decided—a model with low overall state-reconstruction error can still be useless at the boundary, which is the only region the critic acts on.

4.3 The safety critic

At deployment, a higher-level policy (a phase-based controller, or later a VLA policy) proposes an assistance action a_t^{pol} . The critic accepts it only if the predicted margin stays above a buffer ϵ over the horizon; otherwise it returns the closest admissible action. Define the worst predicted margin of a candidate a ,

$$b_{\min}(a) = \min_{k=1, \dots, K} \hat{b}_{t+k}(F_\theta, a), \quad (7)$$

and the *predicted-safe set*

$$\mathcal{A}_{\text{safe}}(s_t) = \{a \in \mathcal{A} : b_{\min}(a) \geq \epsilon\}. \quad (8)$$

Algorithm 1 Aegis safety-critic step (receding horizon)

Require: observation s_t ; policy proposal a_t^{pol} ; baseline a_t^{base} ; buffer ϵ ; horizon K ; candidates M

- 1: $z_t \leftarrow E_\psi(s_t)$
- 2: $\mathcal{C} \leftarrow \{a_t^{\text{pol}}, a_t^{\text{base}}\} \cup \{M \text{ samples around } a_t^{\text{pol}}\}$
- 3: **for** $a \in \mathcal{C}$ **do**
- 4: $\hat{z} \leftarrow z_t$; $b_{\min}(a) \leftarrow +\infty$
- 5: **for** $k = 1$ to K **do** $\triangleright$ roll out in latent space
- 6: $\hat{z} \leftarrow F_\theta(\hat{z}, a_k)$ $\triangleright a_1 = a$, then revert to baseline
- 7: $b_{\min}(a) \leftarrow \min(b_{\min}(a), G_\chi(\hat{z}))$
- 8: **end for**
- 9: **end for**
- 10: $\mathcal{A}_{\text{safe}} \leftarrow \{a \in \mathcal{C} : b_{\min}(a) \geq \epsilon\}$
- 11: **if** $\mathcal{A}_{\text{safe}} \neq \emptyset$ **then**
- 12: $a_t^* \leftarrow \arg \min_{a \in \mathcal{A}_{\text{safe}}} \|a - a_t^{\text{pol}}\|_2$
- 13: **else**
- 14: $a_t^* \leftarrow a_t^{\text{base}}$; trigger reaction-wheel recovery
- 15: **end if**
- 16: **return** a_t^*

The deployed action is the policy proposal projected onto the safe set,

$$a_t^* = \arg \min_{a \in \mathcal{A}_{\text{safe}}(s_t)} \|a - a_t^{\text{pol}}\|_2^2, \quad (9)$$

and if $\mathcal{A}_{\text{safe}}(s_t) = \emptyset$ the critic falls back to the certified low-level controller a_t^{base} and commands a reaction-wheel recovery. If the proposal is already safe, (9) returns it unchanged. Algorithm 1 summarizes one critic step; candidates are drawn around a_t^{pol} and always include a_t^{pol} and a_t^{base} .

4.4 Margin guarantee under bounded prediction error

We now show that protecting the *predicted* margin protects the *true* margin, provided the buffer dominates the model’s error. We first state the error budget.

Assumption 1 (Bounded prediction error). Along the realized trajectory, for each horizon step k the world model’s ground-plane CoM position and velocity errors are bounded,

$$\|\hat{p}_{c,t+k}^{xy} - p_{c,t+k}^{xy}\| \leq \delta_k, \quad \|\hat{\dot{p}}_{c,t+k}^{xy} - \dot{p}_{c,t+k}^{xy}\| \leq \delta'_k,$$

and the predicted support polygon edges are within Hausdorff distance ρ_k of the true edges. The margin head is consistent with the decoded geometry, $\hat{b}_{t+k} = \min_{e \in \partial \hat{\mathcal{S}}_{t+k}} \text{dist}(\hat{\xi}_{t+k}, e)$ up to a tolerance absorbed into ρ_k .

Lemma 1 (Margin is 1-Lipschitz in the XCoM and polygon). *For a fixed polygon, $b = \min_e \text{dist}(\xi, e)$ satisfies $|b(\xi) - b(\xi')| \leq \|\xi - \xi'\|$, since distance to a fixed set is 1-Lipschitz and the pointwise minimum of 1-Lipschitz functions is 1-Lipschitz. Moreover, displacing every edge by at most ρ changes b by at most ρ .*

Proposition 1 (Predicted-safe implies true-safe). *Let $\Delta_k = \delta_k + \delta'_k/\omega_0 + \rho_k$. If the safety buffer satisfies $\epsilon \geq \max_{k \leq K} \Delta_k$, then any action a with $b_{\min}(a) \geq \epsilon$ keeps the true margin nonnegative over the horizon: $b_{t+k} \geq 0$ for all $k \leq K$. In particular, the true extrapolated center of mass remains within the support polygon for the entire horizon under the deployed action a_t^* from (9).*

Proof. By (3) the XCoM is an affine function of CoM position and velocity, so the predicted XCoM error obeys

$$\|\hat{\xi}_{t+k} - \xi_{t+k}\| \leq \|\hat{p}_{c,t+k}^{xy} - p_{c,t+k}^{xy}\| + \frac{1}{\omega_0} \|\hat{\dot{p}}_{c,t+k}^{xy} - \dot{p}_{c,t+k}^{xy}\| \leq \delta_k + \delta'_k/\omega_0.$$

By Lemma 1, combining the XCoM displacement with the at-most- ρ_k polygon displacement gives $|\hat{b}_{t+k} - b_{t+k}| \leq \delta_k + \delta'_k/\omega_0 + \rho_k = \Delta_k$. Hence $b_{t+k} \geq \hat{b}_{t+k} - \Delta_k$. For an action with $b_{\min}(a) \geq \epsilon$ we have $\hat{b}_{t+k} \geq \epsilon$ for all $k \leq K$, so

$$b_{t+k} \geq \epsilon - \Delta_k \geq \epsilon - \max_{j \leq K} \Delta_j \geq 0.$$

Since $a_t^* \in \mathcal{A}_{\text{safe}}(s_t)$ whenever the set is nonempty, the true margin stays nonnegative and $\xi_{t+k} \in \mathcal{S}_{t+k}$ over the horizon. $\square$

Remark 1 (Why this is falsifiable). Proposition 1 is not a claim that the model is accurate; it is a conditional contract. It tells us *exactly* what to measure—the per-horizon errors $\delta_k, \delta'_k, \rho_k$ —and exactly how to set the buffer. If, on held-out perturbation trials, the realized errors exceed the budget used to set ϵ , the guarantee is violated and the claim is refuted. This is the testable statement the evaluation in Sec. 6 is designed to probe.

In practice Δ_k is unknown a priori and the dynamics may be stochastic. We obtain an operational buffer by split-conformal calibration [22]: on a held-out calibration set of trajectories we compute the nonconformity scores $r_k^{(i)} = |\hat{b}_{t+k}^{(i)} - b_{t+k}^{(i)}|$ and set Δ_k to the $\lceil (1 - \alpha)(n + 1) \rceil$ -th smallest score per horizon step.

Corollary 1 (Conformal margin guarantee). *Under exchangeability of the calibration and test trajectories, if $\epsilon \geq \max_{k \leq K} \Delta_k$ with Δ_k the split-conformal quantile at level $1 - \alpha$, then for a test state the deployed action satisfies $\mathbb{P}(b_{t+k} \geq 0, \forall k \leq K) \geq 1 - \alpha$ (per-step, with a union bound over k giving $1 - K\alpha$ jointly; a Bonferroni or max-score calibration tightens this).*

4.5 Training objective

We train $E_\psi, F_\theta, D_\phi, G_\chi$ jointly on K -step rollouts with a latent-consistency term, a state-reconstruction term, a contact-classification term, and the boundary-weighted margin term:

$$\mathcal{L} = \sum_{k=1}^K \left[\lambda_z \|\hat{z}_{t+k} - \text{sg}(E_\psi(s_{t+k}))\|^2 + \lambda_r \|\hat{s}_{t+k} - s_{t+k}\|^2 + \lambda_c \text{BCE}(\hat{c}_{t+k}, c_{t+k}) + \lambda_b w(b_{t+k}) (\hat{b}_{t+k} - b_{t+k})^2 \right], \quad (10)$$

where $\text{sg}(\cdot)$ is the stop-gradient, BCE is binary cross-entropy on foot contacts, and the boundary weight $w(b) = 1 + \beta \exp(-b^2/2\sigma_b^2)$ upweights states near $b=0$. Upweighting the boundary is what makes the calibrated Δ_k small in the safety-critical region.

4.6 Deployment and the control hierarchy

Aegis is the middle of a strict rate hierarchy (Fig. 3). A high-level policy runs at 1–10 Hz and sets *what* to do; the Aegis critic runs at 10–50 Hz as the fall-risk gate; a mid-level controller emits a terrain-conditioned torque reference at 100–200 Hz; and a certified joint impedance/torque controller closes the loop at 1 kHz with hard saturation limits. Critically, the bottom two layers are classical and certifiable and never depend on a neural network being available or correct: if the world model is uncertain or $\mathcal{A}_{\text{safe}} = \emptyset$, control degrades gracefully to the low-level controller, never to a neural guess. Expressing assistance as a *delta* over the nominal controller (Sec. 3.2) keeps the action distribution small and makes the projection (9) well-conditioned.

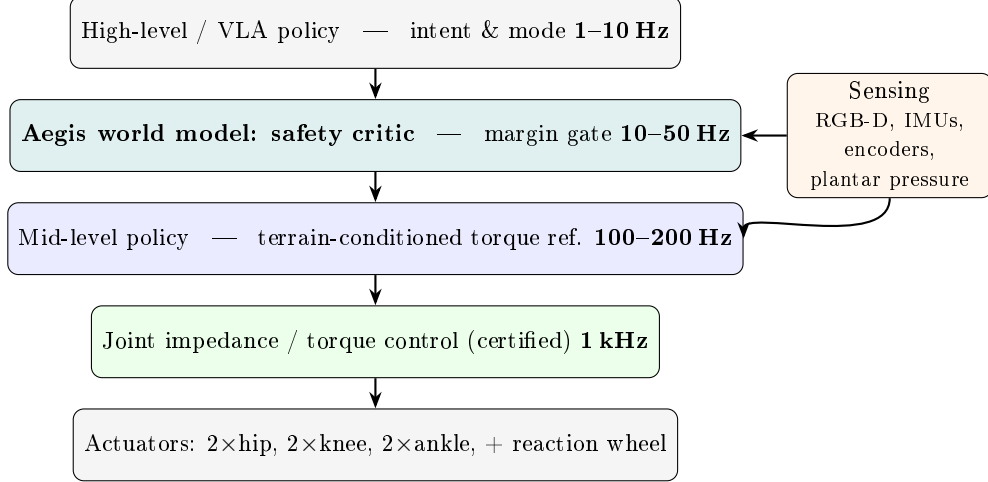


Figure 3: Control hierarchy. The slow, advisory policy sits on top; Aegis is the fall-risk gate; the two fast bottom layers are classical and certifiable. Sensing feeds the state estimate consumed by the critic and the mid-level controller. The big model is intermittent by design—this is a feature of the safety-critic framing, not a limitation to hide.

5 Frontal-Plane Authority of the Reaction Wheel

We now make precise the claim that the pelvis reaction wheel is not a luxury but a controllability requirement in single support. We use a reduced-order frontal-plane model and show the wheel strictly enlarges the one-step recoverable set.

Reduced-order model. Let y be the lateral CoM position, $z_c = \ell$ the (constant) CoM height, m the mass, and $\omega_0 = \sqrt{g/z_c}$. The lateral center of pressure p is constrained to the admissible region $p \in [p^-, p^+]$, which is the lateral extent of the support polygon: *narrow* in single support, *wide* in double support. Following the linear inverted pendulum with a centroidal moment [4, 8], the lateral dynamics with reaction-wheel moment τ_{rw} are

$$\ddot{y} = \omega_0^2 (y - p) - \frac{1}{m z_c} \tau_{\text{rw}}, \quad |\tau_{\text{rw}}| \leq \bar{\tau}_{\text{rw}}. \quad (11)$$

With the lateral XCoM $\xi = y + \dot{y}/\omega_0$, differentiating gives the first-order XCoM dynamics

$$\dot{\xi} = \omega_0 (\xi - p) - \frac{1}{m z_c \omega_0} \tau_{\text{rw}}. \quad (12)$$

Suppose the body is leaning laterally toward the outer edge p^+ so the relevant margin is $b = p^+ - \xi \geq 0$. To increase the margin we make $\dot{\xi}$ as negative as possible. The CoP term is most helpful at $p = p^+$, and the wheel term is most helpful at $\tau_{\text{rw}} = +\bar{\tau}_{\text{rw}}$.

Proposition 2 (The reaction wheel strictly enlarges the recoverable set). *Fix a step $\Delta t > 0$ and buffer $\epsilon \geq 0$. Define the one-step recoverable set as the set of lateral states from which some admissible control keeps the margin above ϵ ,*

$$\mathcal{R} = \{(\xi, \dot{\xi}) : \exists p \in [p^-, p^+], \tau_{\text{rw}} \in [-\bar{\tau}_{\text{rw}}, \bar{\tau}_{\text{rw}}] \text{ s.t. } b_{t+1} \geq \epsilon\}.$$

Let $\mathcal{R}_{\text{foot}}$ be this set with $\bar{\tau}_{\text{rw}} = 0$ (foot-only) and $\mathcal{R}_{\text{foot}+\text{rw}}$ with $\bar{\tau}_{\text{rw}} > 0$. Then, whenever the wheel is not speed-saturated, $\mathcal{R}_{\text{foot}} \subsetneq \mathcal{R}_{\text{foot}+\text{rw}}$, and the most negative achievable XCoM rate improves by exactly

$$\Delta \dot{\xi} = \frac{\bar{\tau}_{\text{rw}}}{m z_c \omega_0} > 0.$$

Proof sketch (full proof in App. A). From (12), with $p = p^+$ the foot-only minimum rate is $\dot{\xi}_{\text{foot}}^{\min} = \omega_0(\xi - p^+)$. Adding $\tau_{\text{rw}} = +\bar{\tau}_{\text{rw}}$ gives $\dot{\xi}_{\text{foot}+\text{rw}}^{\min} = \omega_0(\xi - p^+) - \bar{\tau}_{\text{rw}}/(m z_c \omega_0)$, which is strictly smaller by $\Delta \dot{\xi}$. A more negative attainable $\dot{\xi}$ means the post-step margin $b_{t+1} \approx b_t - \dot{\xi} \Delta t$ is attainable at a strictly larger set of initial ξ , so $\mathcal{R}_{\text{foot}} \subseteq \mathcal{R}_{\text{foot}+\text{rw}}$ with strict inclusion on the boundary layer of width $\Theta(\Delta \dot{\xi} \Delta t)$. In single support p^+ is small (narrow stance), so when $\xi > p^+$ the foot term $\omega_0(\xi - p^+)$ is *positive*—it drives the XCoM further out—and the wheel term is the only source of inward authority, which is the regime where $\mathcal{R}_{\text{foot}}$ can be empty while $\mathcal{R}_{\text{foot}+\text{rw}}$ is not. $\square$

Remark 2 (Transient authority). The wheel saturates in speed, so the inward authority $\Delta \dot{\xi}$ is available only for a bounded duration before the wheel must be unwound. This matches the intended use—brief push/stumble recovery over the same 0.5–1 s horizon the world model predicts—rather than sustained postural support, for which the legs remain responsible. Proposition 2 therefore says the wheel is what makes $\mathcal{A}_{\text{safe}}$ non-empty in the moments that decide a fall, complementing, not replacing, the leg actuators.

6 Experimental Design and Evaluation Protocol

We describe the evaluation we will run. Tables 1–3 give the reporting format; *the numerical entries are illustrative placeholders to demonstrate the format and are to be replaced with measured values.*

6.1 Simulation environment and data

We build a rigid-body model of the wearer plus exoskeleton in a physics simulator [26] (with an Isaac-based variant for parallelism [27]), instrumented to expose the state of (1), foot-contact geometry for $\mathcal{S}_t$, and ground-truth CoM for the margin of (4). Training data comprise locomotion over varied terrain (flat, ramps, stairs, compliant ground) with *lateral and sagittal push perturbations* sampled in magnitude, direction, and gait phase, plus domain randomization over mass, inertia, contact, and sensor noise. The terrain heightmap H_t is rendered from an egocentric depth camera pose. We hold out perturbation trials for conformal calibration (Cor. 1) and for closed-loop testing.

6.2 Baselines

- **Reactive-Threshold:** applies a fixed corrective profile once a proprioceptive instability threshold trips (no prediction).
- **Blind-WM:** the Aegis critic with proprioception only (H_t removed)—isolates the value of exteroception.
- **LSTM-FallPredictor:** a recurrent classifier predicting fall/no-fall from IMU, used as a gate; no action-conditioned rollout.
- **MPC-Oracle:** model-predictive control with access to the (near) true dynamics—an upper-reference for the critic’s role, not a deployable method.

Table 1: Closed-loop balance under held-out perturbations. **Illustrative placeholder values to demonstrate the reporting format; to be populated from the experiments of Sec. 6.** Arrows show preferred direction.

Method	Fall rate ↓	Near-fall ↓	Recovery ↑	Interv. rate	Track. RMS ↓
Reactive-Threshold	0.18	0.34	0.61	—	0.21
LSTM-FallPredictor	0.14	0.29	0.66	0.12	0.20
Blind-WM	0.11	0.25	0.71	0.15	0.18
Aegis—RW (no wheel)	0.08	0.19	0.78	0.17	0.17
Aegis (ours)	0.04	0.11	0.89	0.16	0.16
MPC-Oracle (<i>ref.</i>)	0.03	0.09	0.91	—	0.15

Table 2: Predictive quality of the world model. **Illustrative placeholders; to be populated from experiments.** Margin errors in cm; lead time in ms.

Method	CoM err. @0.5 s ↓	Margin MAE ↓	Margin MAE (bdry) ↓	Fall AUROC ↑	Lead time ↑
LSTM-FallPredictor	—	—	—	0.84	180
Blind-WM	4.1	1.9	2.6	0.88	240
Aegis—RW	3.4	1.5	1.9	0.91	300
Aegis	3.2	1.4	1.6	0.94	340

- **Aegis—RW**: our method with the reaction wheel disabled—isolates the hardware contribution of Sec. 5.
- **Aegis (ours)**: full model and critic.

6.3 Metrics

Balance: *fall rate* and *near-fall rate* (fraction of trials with $\min_t b_t < 0$, resp. $< b_{\text{near}}$) under held-out pushes; *recovery success* after a standardized impulse. Predictive quality: CoM position error vs. horizon k ; margin MAE *overall* and *near boundary* ($|b| < b_0$); fall-prediction AUROC and *lead time* (how early $\hat{b}$ crosses ϵ before a true $b < 0$). Assistance quality: torque-tracking RMS of the wearer’s intended motion and assistance smoothness. Critic behavior: *intervention rate* (fraction of steps where (9) alters the proposal) and *empty-safe-set rate*.

6.4 Ablations

We ablate (i) exteroception (H_t on/off), (ii) the reaction wheel (Sec. 5), (iii) boundary weighting in (10) (β), (iv) horizon K , and (v) the buffer ϵ and its conformal calibration. We expect exteroception and boundary weighting to improve near-boundary margin accuracy and lead time, the wheel to lower the empty-safe-set rate in single support (testing Prop. 2), and the conformal ϵ to trade intervention rate against the realized fall rate in the way Prop. 1 predicts.

6.5 Hardware validation plan

On the physical device (Fig. 1) we will (a) quantify the CoM-estimation error against motion capture—the guarantee in Prop. 1 rests on this estimate, so we report it as a primary number; (b) measure the per-horizon errors $\delta_k, \delta'_k, \rho_k$ on treadmill perturbation trials and verify the conformal

Table 3: Ablations. **Illustrative placeholders; to be populated from experiments.** “Empty-safe” is the empty-safe-set rate in single support.

Configuration	Fall rate ↓	Margin MAE (bdry) ↓	Lead time ↑	Empty-safe ↓
Full Aegis	0.04	1.6	340	0.02
– exteroception (H_t)	0.07	2.4	250	0.03
– reaction wheel	0.08	1.7	320	0.09
– boundary weighting	0.06	2.5	280	0.04
horizon K : 1.0→0.3 s	0.09	1.8	150	0.05
buffer $\epsilon=0$ (no margin)	0.13	1.6	–	0.00

coverage of Cor. 1; (c) run standardized cable-pull push-recovery trials with the wheel enabled vs. disabled to test Prop. 2; and (d) confirm graceful degradation to the certified low-level controller when the model is withheld. All hardware testing follows the device’s certified torque and watchdog limits, with the 1 kHz controller authoritative at all times.

7 Discussion, Limitations, and the Path to a VLA

The bound is conditional—honestly. Proposition 1 assumes a bounded prediction error; the deterministic core and the conformal calibration make this operational, but the coverage of Cor. 1 is only as valid as the exchangeability between calibration and deployment. Distribution shift (a new wearer, unseen terrain) can break it. We therefore treat conformal coverage as a *monitored* quantity and keep the certified low-level controller as the ultimate fallback; we also report results with a stochastic recurrent state-space variant of F_θ , for which the margin head predicts a quantile and the buffer is set from the predictive distribution.

CoM estimation is the long pole. Every guarantee flows through p_c . A biased estimate makes ϵ meaningless, so we budget real effort on kinematic-plus-IMU fusion and report CoM error as a first-class metric rather than folding it silently into ρ_k .

Latency and onboard compute. A large model cannot run in the 1 kHz loop; the rate hierarchy (Fig. 3) assumes the critic is intermittent (10–50 Hz) and the policy slower still. This is a strength of the safety-critic framing—the heavy model is advisory and clippable—and we foreground it rather than hide it.

From critic to VLA policy. Aegis is the world-model substrate for a downstream language-conditioned VLA policy [23, 24]: the VLA proposes assistance from an instruction and the scene (“help me up these stairs,” “walk slower on this ice”), and the *same* critic of Sec. 4.3 filters its proposals through (9). The world model thus does double duty—training-time imagination for the policy and deployment-time safety for both. We see this as the natural second step, with the present paper establishing the action-conditioned latent world model and its margin guarantee as the foundation.

8 Conclusion

We presented Aegis, an action-conditioned latent world model that serves as a margin-of-stability safety critic for a lower-limb exoskeleton with a pelvis reaction wheel. By predicting the XCoM-based margin under candidate assistance torques and projecting unsafe proposals onto a predicted-safe set, Aegis turns “do not destabilize the wearer” into a checkable, calibrated constraint with a falsifiable guarantee, and it shows that a specific hardware choice—the reaction wheel—is what keeps that constraint feasible in the single-support moments when falls happen. The evaluation protocol is designed to test the guarantee rather than merely to report accuracy, and the framework is built to carry a language-conditioned policy on top without weakening its safety contract.

A Full proof of Proposition 2

Consider the lateral XCoM dynamics (12) with admissible CoP $p \in [p^-, p^+]$ and wheel torque $\tau_{\text{rw}} \in [-\bar{\tau}_{\text{rw}}, \bar{\tau}_{\text{rw}}]$. The attainable set of XCoM rates at a state ξ is the interval

$$\dot{\xi} \in \left[\omega_0(\xi - p^+) - \frac{\bar{\tau}_{\text{rw}}}{mz_c\omega_0}, \quad \omega_0(\xi - p^-) + \frac{\bar{\tau}_{\text{rw}}}{mz_c\omega_0} \right],$$

since (12) is affine and monotone in both p (decreasing) and τ_{rw} (decreasing), so the extremes are attained at the box corners. With $\bar{\tau}_{\text{rw}} = 0$ the lower endpoint is $\omega_0(\xi - p^+)$; with $\bar{\tau}_{\text{rw}} > 0$ it is smaller by $\Delta\dot{\xi} = \bar{\tau}_{\text{rw}}/(mz_c\omega_0) > 0$. Using the first-order margin update $b_{t+1} = b_t - \dot{\xi}\Delta t + o(\Delta t)$ toward the edge p^+ , the condition $b_{t+1} \geq \epsilon$ is feasible iff the attainable $\dot{\xi}$ can be made at most $(b_t - \epsilon)/\Delta t$. Because the attainable lower endpoint is strictly smaller with the wheel, the set of ξ satisfying feasibility grows; the additional recoverable states are exactly those with $(b_t - \epsilon)/\Delta t \in [\omega_0(\xi - p^+) - \Delta\dot{\xi}, \omega_0(\xi - p^+)]$, a layer of width $\Theta(\Delta\dot{\xi}\Delta t)$. When $\xi > p^+$ (single support, narrow stance, body past the outer foot edge) the foot-only lower endpoint $\omega_0(\xi - p^+)$ is positive, so no foot-only control yields $\dot{\xi} \leq 0$ and $\mathcal{R}_{\text{foot}}$ is empty for $\epsilon > b_t$, whereas $\mathcal{R}_{\text{foot}+\text{rw}}$ is non-empty as long as $\Delta\dot{\xi}$ exceeds $\omega_0(\xi - p^+) - (b_t - \epsilon)/\Delta t$ and the wheel is unsaturated. $\square$

B Default hyperparameters and architecture

Latent dimension $d=256$; encoder: 3-layer MLP for proprio/centroidal, 4-layer CNN for H_t ($32 \times 32 \rightarrow 64$ -d), concatenated; dynamics F_θ : residual GRU core (2×256); decoder/heads: 2-layer MLPs; horizon $K=25$ at $\Delta t=40$ ms ($T_H=1.0$ s). Loss weights $\lambda_z=1, \lambda_r=1, \lambda_c=0.5, \lambda_b=5$; boundary weight $\beta=4, \sigma_b=3$ cm. Conformal level $\alpha=0.05$ per step. Optimizer AdamW, lr 3×10^{-4} , batch 256 roll-outs. Candidates $M=16$ sampled as Gaussian perturbations of a^{pol} plus the baseline. (*Values are sensible defaults to be tuned.*)

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